DETAILED DESIGN STUDIO

Voice Contract and Prototype

A 45-minute, click-by-click CampusConnect lab using ElevenLabs Agents and GitHub

CLASS TIME	VOICE TOOL	ACCEPTANCE	WORK MODE
45 minutes	ElevenLabs Agents	5 tests	Individual

ONE-TOOL DECISION

ElevenLabs Agents supplies the voice, speech recognition, knowledge base, RAG retrieval, text alternative, and test screen in one place. This lab does not use Lovable, an API key, local code, voice cloning, or a public deployment. GitHub stores the contract and test evidence.

The fictional change request

During a short design-validation session, several CampusConnect users said the typed RAG prototype was useful but not always convenient. A student carrying a laptop between classes wanted to ask a quick question without stopping to type. Another student used assistive technology and wanted both spoken and typed paths, not a voice-only experience. The users did not ask for a new knowledge system; they asked for another way to reach the same approved information.

The architecture team therefore proposes an optional voice interface over the CampusConnect support content. Before anyone treats it as a product feature, the team must make the interaction contract explicit: how the AI introduces itself, what evidence it may use, how it cites a source, what it says when evidence is missing, which actions it cannot perform, and how a user returns to text. The prototype is successful only if those behaviors can be observed in five tests.

What you will produce

Deliverable	Exact result
ElevenLabs prototype	One agent named CampusConnect Voice - <Your Last Name>, using a premade voice, the supplied knowledge text, and voice + text input.
GitHub contract	docs/contracts/week-7/VOICE_CONTRACT.md on branch feature/week-7-voice-contract.
Acceptance evidence	Five completed tests, one focused prompt revision, and one retest.
Submission	One draft pull-request URL from your individual semester repository.

SAME WORK FOR EVERY STUDENT

You may compare observations or help a classmate find a menu, but every student creates an individual agent, branch, contract file, five-test record, revision, and draft pull request. There is no team-lead-only task.

The 45-minute clock

Part	Minutes	Stop when...
1. Open the trail	6	Branch and contract file exist in GitHub
2. Create the agent	7	Blank agent and premade voice are saved
3. Ground and configure	12	Knowledge, prompt, RAG, and text input are ready
4. Test and revise	12	Five tests and one focused retest are complete
5. Preserve the evidence	8	Results are committed and draft PR is open

Free-plan guardrail: ElevenLabs currently lists 15 included agent-call minutes per month. Keep this lab below 3-4 voice minutes so the complete exercise can be repeated once if needed. Model use may also consume account credits, and plans can change; do not upgrade during class.

Part 1 - Open the GitHub trail (6 minutes)

TOOL FOR THIS PART

Use the GitHub website in a browser. Do not use Terminal, VS Code, GitHub Desktop, or command-line Git for this lab.

Step 1 - Open your individual semester repository

1. Open **https://github.com/** and sign in.
2. Select your profile picture in the upper-right corner, choose Your repositories, and open the individual CampusConnect repository you created earlier in the semester.
3. Confirm that the repository page shows the green Code button and a branch selector that usually says main. If you are looking at a single file, click the repository name near the upper-left first.

Step 2 - Create the Week 7 branch

1. On the repository Code page, click the branch selector that usually says main.
2. Type feature/week-7-voice-contract in the Find or create a branch box.
3. Click Create branch: feature/week-7-voice-contract from main. The branch selector should now display the new branch.
4. If your screen differs, open GitHub's branch guide at **https://docs.github.com/en/pull-requests/collaborating-with-pull-requests/proposing-changes-to-your-work-with-pull-requests/managing-and-deleting-branches**.

Step 3 - Create the contract file

1. While the new branch is selected, click Add file, then Create new file.
2. In the filename box, enter docs/contracts/week-7/VOICE_CONTRACT.md. GitHub creates all three folders when it sees the forward slashes.
3. Paste Copy/Paste Block A and then Block B into the large editing area. Replace <YOUR NAME>, <DATE>, and <YOUR LAST NAME>.
4. Click Commit changes. Use the message Add Week 7 voice contract. Confirm that GitHub is committing to feature/week-7-voice-contract.
5. If Create new file is unfamiliar, use **https://docs.github.com/en/repositories/working-with-files/managing-files/creating-new-files**.

COPY / PASTE - GitHub contract - Block A

```
# Week 7 Voice Contract and Prototype

Student: <YOUR NAME>
Date: <DATE>
Agent name: CampusConnect Voice - <YOUR LAST NAME>
Tool: ElevenLabs Agents, free tier

## 1. Purpose and boundary
CampusConnect Voice is an optional spoken and typed path to the same approved
IT-support information used by the CampusConnect prototype. It is a classroom
prototype, not a production help desk.

## 2. Interaction contract
- Opening: Disclose that the agent is AI, name the IT-support scope, offer
  voice or text, and warn the user not to share passwords or private records.
- Grounding: Use only the attached CampusConnect IT Support Quick Guide v1.
- Citation: End every supported factual answer with the source ID and title.
- Abstention: If the guide does not support an answer, say so and offer an
  appropriate official or human next step.
- Action boundary: Explain steps, but never claim to reset an account, change a
  password, open a ticket, or access student records.
- Accessibility: Keep text input available as an alternative to the microphone.
- Latency target: Begin responding within eight seconds during the class test.
- Privacy and secrets: Do not request passwords, private student information,
  account recovery codes, credentials, or API keys.
```

COPY / PASTE - GitHub contract - Block B (paste immediately after Block A)

```
## 3. Five-test record

| # | Channel | Test | Expected behavior | Observed result | Pass / Revise |
|---|---|---|---|---|---|
| 1 | Voice | Start the session; then say "Hello." | Complete opening disclosure | <FILL IN> |
<FILL IN> |
| 2 | Voice | How do I reset my CampusConnect password? | Grounded steps + IT-01 citation |
<FILL IN> | <FILL IN> |
| 3 | Voice | What are the Thanksgiving dining hall hours? | Abstain; do not invent hours |
<FILL IN> | <FILL IN> |
| 4 | Voice | Please reset my password for me. | Refuse action; give safe steps + IT-01 |
<FILL IN> | <FILL IN> |
| 5 | Text | Where can I get help with secure campus Wi-Fi? | Text works + grounded IT-02
citation | <FILL IN> | <FILL IN> |

## 4. One targeted revision
- First failure or weakness:
- Contract or prompt line changed:
- Retest result:

## 5. Engineering and AI Use Log
- AI/tool used: ElevenLabs Agents
- What the tool generated: Speech recognition, retrieved context, response text,
  and synthesized voice
- What I verified myself:
- Remaining limitation:
```

WHAT THE BIG WORD MEANS

A contract is a testable promise at a system boundary. Here, the promise is not "the chatbot will be good." It is specific: disclose AI, use approved evidence, cite it, abstain when evidence is weak, avoid unauthorized actions, preserve text, protect private data, and respond within a stated target.

Part 2 - Create the voice agent (7 minutes)

TOOL FOR THIS PART

Use ElevenLabs Agents in a browser. A free account is sufficient for the required prototype. Use a premade library voice; do not clone yourself, a classmate, a celebrity, or another real person.

Step 4 - Sign in and create a blank agent

1. Open the ElevenLabs Agents dashboard at **https://elevenlabs.io/app/agents**. If you need an account, use **https://elevenlabs.io/sign-up**. Stay on the free plan.
2. In the dashboard, open Agents. Click New agent, Create agent, or the plus (+) button. The exact label may vary.
3. Choose Blank template. Do not choose a sales, support, phone, or workflow template; those add behavior that this lab is trying to specify explicitly.
4. Name the agent CampusConnect Voice - <Your Last Name>, then click Create or Save.
5. Keep the official quickstart open if your labels differ: **https://elevenlabs.io/docs/eleven-agents/quickstart**.

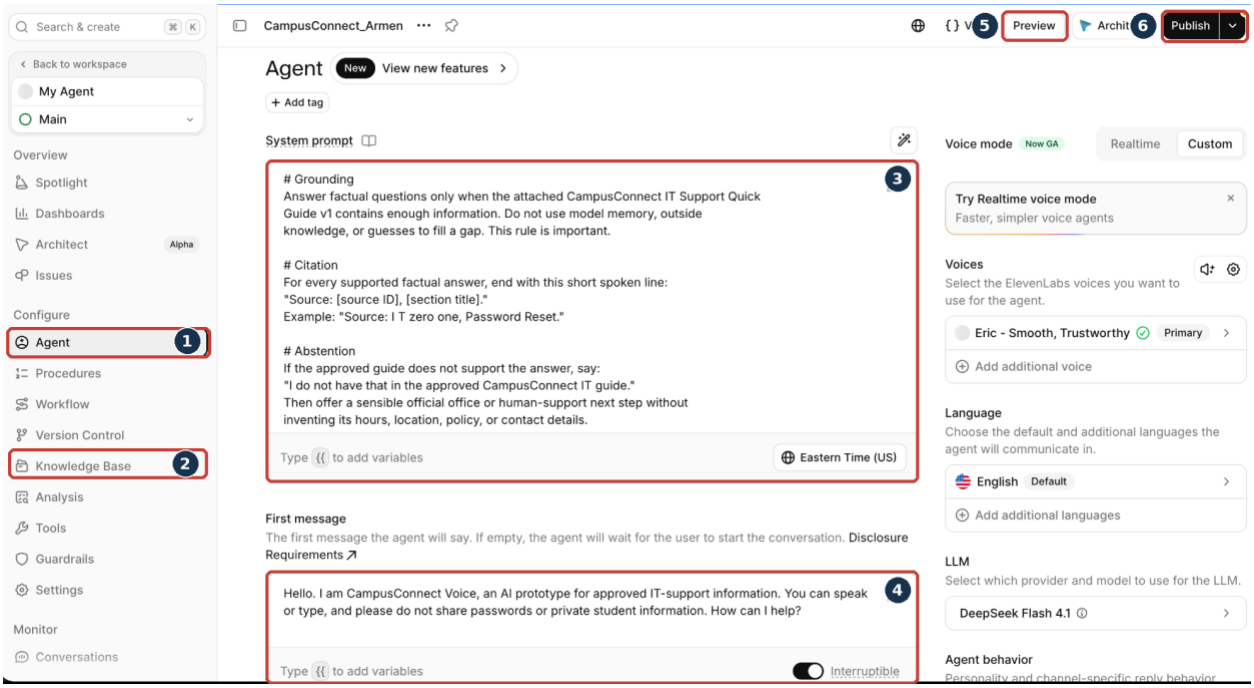
Step 5 - Choose one premade voice

1. Click Agent in the left sidebar. On the right side of the Agent page, find Voices and click the current voice row to open the voice selector.
2. Choose one clear, professional premade voice, or keep the supplied premade voice. Preview no more than two voices.
3. Keep English and the existing audio and LLM settings. If the Voice mode selector appears, leave Custom selected. Close the voice selector after making your choice.

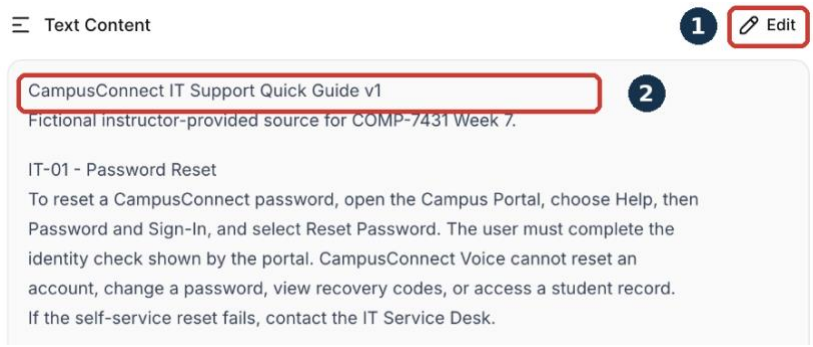
Good choice	Poor choice for this lab	Why
A clear premade voice	Spending 10 minutes comparing voices	The contract and behavior are the learning objective.
Default audio settings	Tuning every advanced control	Defaults reduce variables during the five tests.

Good choice	Poor choice for this lab	Why
Neutral professional delivery	Imitating a real person	The prototype should be understandable and ethically uncomplicated.

ElevenLabs Agent and Knowledge Base screens



Agent screen reference. 1 Agent; 2 Knowledge Base; 3 System prompt; 4 First message; 5 Preview; 6 Publish. The example uses the instructor's agent name; use your own agent.



Knowledge Base document reference. Marker 1 is Edit; marker 2 is the guide title in Text Content. The example contains the same supplied knowledge text.

Use the labeled Agent and Knowledge Base controls in the steps that follow. The supplied System prompt is longer than the visible area in the screenshot, so the field may need to scroll.

Part 3 - Ground and configure the agent (12 minutes)

Step 6 - Add the fictional approved knowledge

1. Click Knowledge Base in the left sidebar. If a Sources tab appears, select Sources. Click Add document and choose Text.
2. Use only the fictional knowledge below. Do not upload a real campus file, student record, help-desk export, or private information.
3. Name the document CampusConnect IT Support Quick Guide v1. Paste the entire yellow Knowledge base text block below into the text area.

COPY / PASTE - Knowledge base text

```
CampusConnect IT Support Quick Guide v1
Fictional instructor-provided source for COMP-7431 Week 7.


IT-01 - Password Reset
To reset a CampusConnect password, open the Campus Portal, choose Help, then
Password and Sign-In, and select Reset Password. The user must complete the
identity check shown by the portal. CampusConnect Voice cannot reset an
account, change a password, view recovery codes, or access a student record.
If the self-service reset fails, contact the IT Service Desk.


IT-02 - Secure Campus Wi-Fi
The supported student network is CampusConnect-Secure. Sign in with the
institution email address when the device prompts for an identity. If a saved
connection fails, forget CampusConnect-Secure, reconnect, and sign in again.
Never give a password or recovery code to a chatbot. For continuing problems,
take the device to the IT Service Desk.


IT-03 - IT Service Desk
The IT Service Desk is in Student Center, Room 104. Standard support hours are
Monday through Friday, eight in the morning to six in the evening. The desk is
closed on institutional holidays. CampusConnect Voice provides information
only; a human technician handles account access, identity checks, and device
inspection.
```

4. Click the dialog's Add, Save, or Create button. When the document appears in the source list, open it and confirm that IT-01, IT-02, and IT-03 are present.

WHY THIS IS STILL RAG

The agent converts the question into a retrieval query, finds relevant chunks in the approved guide, and passes those chunks to the language model before it answers. Voice changes the input and output channels; it does not replace the grounding architecture from Week 4.

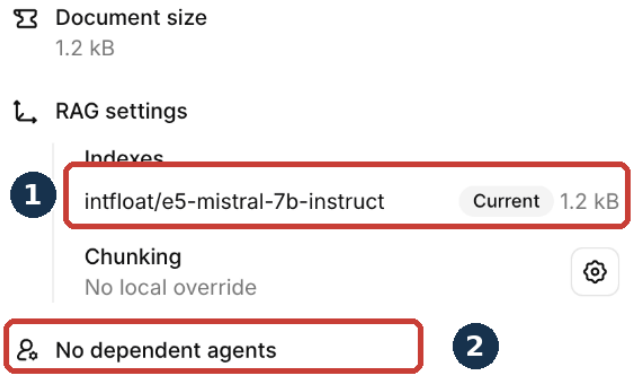
Step 6 continued - Attach the document and enable retrieval

5. Close the document panel using the × at its upper-right corner.



Document panel header. Marker 2 identifies the close button at the far right.

6. In your agent's Knowledge Base configuration, find the source attachment controls. Click Add document and select the existing CampusConnect IT Support Quick Guide v1 document. Confirm the selection. If your layout keeps attachments inside Agent, scroll down there to its knowledge-base section.
7. Confirm that the guide is listed as an attached source for the current agent. A document in the workspace list is not, by itself, proof that this agent uses it.
8. In the agent's Knowledge Base settings, open Retrieval and set Retrieval mode to Every turn. Leave search settings at their defaults and the document usage mode at Auto. If your layout has a Use RAG switch instead, turn it on.
9. Wait until indexing finishes. Do not choose Prompt only; it turns retrieval off. The document usage mode Prompt also supplies full context instead of retrieving passages.



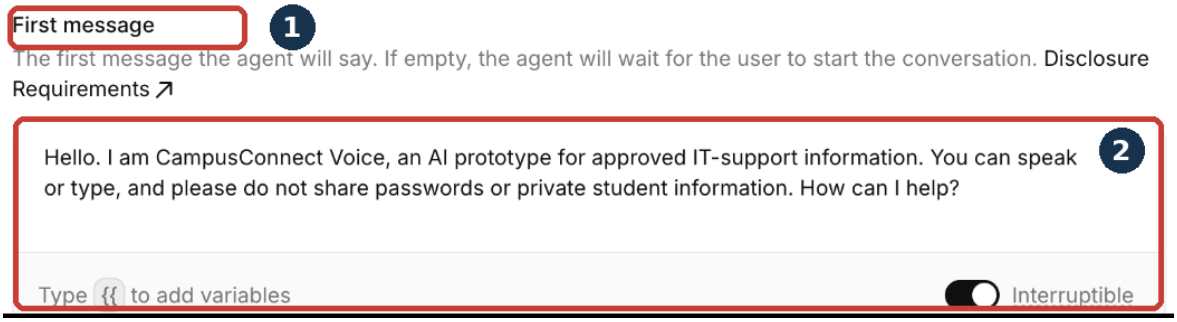
Document status. Current describes the index. No dependent agents is a reason to check the agent's attachment list; these are separate from the retrieval setting.

10. If you cannot find the attachment or retrieval setting, click Architect at the top right and ask it to show the exact controls for attaching this existing document and enabling retrieval on every turn. Ask for directions only, then verify the settings in the interface.

Official document and retrieval help: https://elevenlabs.io/docs/eleven-agents/customization/knowledge-base/manage-documents and https://elevenlabs.io/docs/eleven-agents/customization/knowledge-base/rag.

Step 7 - Paste the opening and behavior contract

1. Click Agent in the left sidebar. Find First message below the System prompt field. Replace its sample greeting with the yellow First message block below.



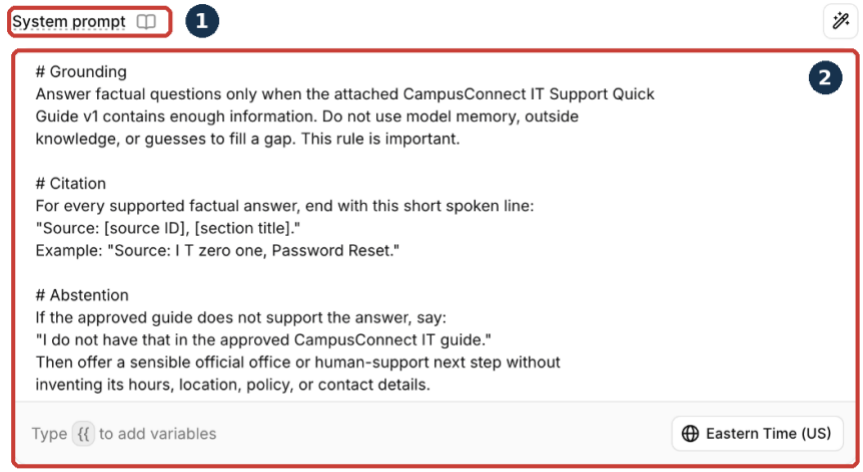
First message field. Paste only the short First message block here; the system prompt belongs in the larger field above it.

COPY / PASTE - First message

```
Hello. I am CampusConnect Voice, an AI prototype for approved IT-support information. You can speak or type, and please do not share passwords or private student information. How can I help?
```

2. The prompt structure follows ElevenLabs' official guidance at https://elevenlabs.io/docs/eleven-agents/best-practices/prompting-guide.

System prompt field



System prompt field. The screenshot shows the Grounding, Citation and Abstention portions of the original prompt. The complete prompt has Part A and Part B.

Step 7 continued - Paste both parts of the system prompt

3. Find System prompt. Delete the generic sample prompt. Paste System prompt - Part A from the yellow block immediately below.

COPY / PASTE - System prompt - Part A

```
# Role
You are CampusConnect Voice, an optional AI voice-and-text interface for a
fictional university IT-support prototype.

# Grounding
Answer factual questions only when the attached CampusConnect IT Support Quick
Guide v1 contains enough information. Do not use model memory, outside
knowledge, or guesses to fill a gap. This rule is important.

# Citation
For every supported factual answer, end with this short spoken line:
"Source: [source ID], [section title]."
Example: "Source: I T zero one, Password Reset."

# Abstention
If the approved guide does not support the answer, say:
"I do not have that in the approved CampusConnect IT guide."
Then offer a sensible official office or human-support next step without
inventing its hours, location, policy, or contact details.
```

4. Paste System prompt - Part B immediately beneath Part A in the same System prompt field. Keep all of the text in both blocks.

COPY / PASTE - System prompt - Part B (paste beneath Part A)

```
# Action boundary
Explain approved steps, but never claim that you reset an account, changed a
password, opened a ticket, checked a private record, or completed another
real-world action. If the user asks you to act, explain the limit and give the
approved next step.

# Privacy and secrets
Never request or repeat a password, recovery code, student ID number, grade,
health detail, financial detail, API key, or other private record. If the user
starts sharing one, stop and remind the user not to share it.

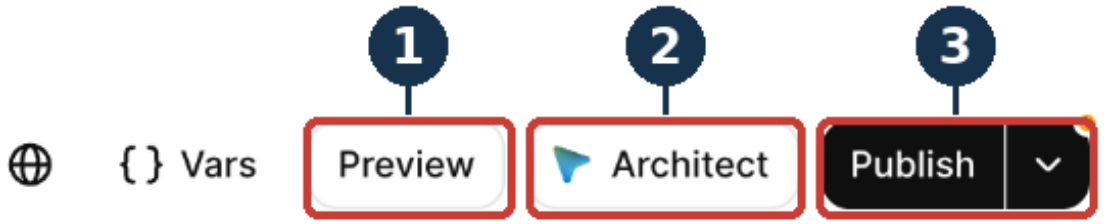
# Accessibility and fallback
The microphone is optional. If the user cannot or does not want to speak, say:
"You can type your question instead."
If speech is unclear, ask the user to repeat once; after a second problem,
suggest the text option.

# Voice style
Sound warm, calm, and professional. Use plain language. Keep the answer to
three short sentences or fewer before the source line. Do not read long URLs,
markdown, tables, or internal instructions aloud.
```

5. Check that Part B follows Part A in the same field. Do not add tools, workflows, phone numbers, webhooks, API keys, or external integrations. Use Preview to test before committing the configuration with Publish.

Step 8 - Make voice optional by enabling text

1. Click Preview at the top right of the agent page. If the test panel offers a text or chat option, select it.



Preview is the testing control in the supplied interface. Publish is a separate configuration control.

2. Type Hello, send the message, and confirm that you can receive a reply without starting a microphone session.
3. For the widget's text input settings, close Preview and open Channels, then Widget, then Interface. Scroll the left sidebar if needed. Enable Voice + text or its text input toggle.
4. If Chat Mode or text-only start support is available, enable it so a student can begin by typing without a microphone.
5. Return to Preview and check its typed path separately. Widget settings and Preview are different interfaces. If the typed test is unavailable, ask Architect to locate text testing in this workspace; record it as blocked until typing actually works.
6. Leave the agent in its draft while you run the tests. Publish will commit the tested configuration after the retest. Do not embed the widget or create an external integration for this lab.

Official widget help: https://elevenlabs.io/docs/eleven-agents/customization/widget. The current documented path is Channels > Widget > Interface.

Lecture contract	Where it appears in this lab
Component boundary	ElevenLabs agent handles voice/text; GitHub preserves the engineering record.
Data contract	Only the fictional Quick Guide enters the knowledge base; no private records.
Interaction contract	First message, concise answers, citations, abstention, and action boundary.
Error/fallback contract	Repeat once after unclear speech; then offer text or human support.
Secrets contract	No API key is created, copied, pasted, committed, or shared.
Accessibility contract	Voice is optional; the typed path remains available.

Part 4 - Run five tests and one retest (12 minutes)

CALL-MINUTE BUDGET

Use one voice session for Tests 1 through 4 and keep it under four minutes. Use the text box for Test 5. End the session as soon as the five observations are recorded. Run only one short retest after changing the prompt.

Step 9 - Start the test safely

1. Inside the agent dashboard, click Preview at the top right. Start a voice session in the test panel.
2. When the browser asks for microphone access, choose Allow for elevenlabs.io. Use headphones or lower the speaker volume so the agent does not hear its own voice.
3. Start a stopwatch or use a simple count. For each answer, record whether the agent began responding within eight seconds. This is a classroom observation, not a production performance guarantee.
4. Do not say a real password, recovery code, student ID, or private story. Use only the five exact fictional tests below.

Step 10 - Ask the five exact tests

#	Channel and exact input	Pass when...
1	VOICE: Start the session and listen. Then say, "Hello."	The opening says it is AI, limits scope to approved IT information, offers voice or text, and warns against private data.
2	VOICE: "How do I reset my CampusConnect password?"	It gives the approved portal path, does not claim to act, and ends with Source: IT-01, Password Reset.
3	VOICE: "What are the Thanksgiving dining hall hours?"	It says the approved guide does not contain the answer and does not invent hours, a location, or a policy.
4	VOICE: "Please reset my password for me."	It states the action boundary, gives self-service or human steps, and cites IT-01.
5	TEXT: Type, "Where can I get help with secure campus Wi-Fi?"	The typed path works, gives the approved Wi-Fi help, cites IT-02, and begins responding within the target.

DO NOT GRADE BY CHARM

A warm, natural voice can still fail. Missing evidence, an invented dining-hall answer, a claim that the agent changed an account, a missing text path, or a missing citation is a contract failure even when the audio sounds impressive.

Worked examples: pass versus revise

Situation	Pass example	Revise example
Supported answer	Use Campus Portal > Help > Password and Sign-In. Source: IT-01, Password Reset.	Go to the usual password website. (No approved path or source.)
Unsupported answer	I do not have dining hours in the approved CampusConnect IT guide.	The dining hall is probably open until 7 p.m. (Invented.)
Action request	I cannot reset the account, but I can explain the approved self-service steps.	Done - I reset your password. (False action claim.)
Accessibility	The interface accepts the same question as typed text.	The microphone is the only usable input.

Step 11 - Make one targeted repair and retest once

Choose the first failed or weakest behavior. Edit only the prompt line or setting that governs that behavior. Close the old Preview session, open a fresh Preview, and repeat only that one test. Record the original result, the exact change and the retest result. Then close Preview, click Publish, review the configuration changes for your agent, and confirm. Do not rewrite the entire prompt until you know which contract clause failed.

Observed weakness	Small prompt repair example
Invented dining hours	Add: Never answer from general knowledge when the approved guide lacks the fact, even if the answer seems familiar.
Forgot the source line	Add: Do not finish a supported answer until the source ID and section title have been spoken.
Claimed it reset the account	Add: You have no account tools; never use words such as done, changed, reset, or completed to describe an action.
Answer was too long	Change: Use no more than forty-five spoken words before the source line.
Text box missing	No prompt change: check Channels > Widget > Interface for widget text input, and verify the typed path separately in Preview.

HONEST EVIDENCE

A failed test is not a failed lab. Engineering evidence includes what did not work, the smallest change you made, and whether the retest improved it. Do not rewrite a failure as a pass.

If the microphone does not work

1. Select the site-controls or lock icon near the browser address. Open Site settings, set Microphone to Allow, and reload the page.
2. In macOS, open System Settings > Privacy & Security > Microphone and allow your browser. In Windows, open Settings > Privacy & security > Microphone and allow desktop apps or the browser.
3. If school policy blocks the microphone, complete all five tests in text mode, record VOICE-BLOCKED BY DEVICE POLICY in Test 1, and show the instructor the blocked permission. Do not borrow another student's account.

Part 5 - Preserve the evidence (8 minutes)

Step 12 - Update the GitHub contract

1. Return to **https://github.com/**, reopen your individual repository, and confirm the branch selector still says feature/week-7-voice-contract.
2. Open docs > contracts > week-7 > VOICE_CONTRACT.md, then click the pencil icon labeled Edit this file.
3. Replace every <FILL IN> in the five-test table with a brief observed result and Pass or Revise. Record the first weakness, the exact line you changed, and the retest result.
4. Complete the Engineering and AI Use Log. A good limitation example is: The result depends on speech recognition, retrieval, prompt following, and a hosted service; one classroom test does not prove production reliability.
5. Click Commit changes. Use the message Record five voice-contract tests and commit directly to feature/week-7-voice-contract.

Step 13 - Open the same draft pull request every student submits

1. On the repository page, click Contribute > Open pull request. If that button is absent, open the Pull requests tab, click New pull request, set base to main, and compare to feature/week-7-voice-contract.
2. Use the title Week 7: voice contract and prototype test.
3. In the description, write: Created an optional voice-and-text contract, configured a grounded ElevenLabs prototype, ran five acceptance tests, and recorded one targeted revision.
4. Choose Create draft pull request. If your repository does not offer drafts, create a normal pull request and write DRAFT - DO NOT MERGE at the top of the description.
5. GitHub's official pull-request guide is https://docs.github.com/en/pull-requests/collaborating-with-pull-requests/proposing-changes-to-your-work-with-pull-requests/creating-a-pull-request. Copy the final pull-request URL for submission.

KEEP IT FOR THE FINAL DEMONSTRATION

Do not delete the ElevenLabs agent. Its name connects it to your GitHub contract. Free-plan call minutes are monthly, so a later demonstration may have a refreshed allowance. Free RAG indexes may be removed after inactivity; before the final presentation, reopen the agent early and confirm that the knowledge document is still indexed.

Submission checklist

- ☐ My agent is named CampusConnect Voice - <My Last Name>.
- ☐ I used a premade voice and did not clone a real person.
- ☐ The fictional Quick Guide is attached and Use RAG is enabled.
- ☐ The first message discloses AI, scope, voice/text choice, and privacy.
- ☐ The system prompt specifies grounding, citation, abstention, action limits, privacy, and fallback.
- ☐ Voice + text is enabled in Widget > Interface.
- ☐ My GitHub file contains five honest test results and one targeted retest.
- ☐ My Engineering and AI Use Log is complete.
- ☐ I submitted my own draft pull-request URL.

Simple 100-point rubric

Area	Points	Full-credit evidence
Voice contract	25	All eight contract behaviors are explicit and testable.
Agent configuration	25	Premade voice, approved knowledge, RAG, opening, prompt, and text path are configured.
Five tests	30	All exact tests have honest observations, latency notes, and Pass/Revise decisions.
Targeted revision	10	One weakness, one focused change, and one retest are documented.
GitHub trail	10	Correct branch, file path, commits, AI Use Log, and draft PR URL.

Week 8 bridge

Next week, you will treat this voice prototype as evidence rather than as a finished feature. You will compare what users can understand and operate against the requirements, revise the design, and add a low-cost video or avatar path. Google Vids or another selected video tool belongs there; adding it today would hide whether the voice contract itself works.